



# NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

## Seventh Semester B.E. (Credit System) Degree Examinations

Supplementary Examinations - December 2025

### 21BT8X40 - BIOFUEL ENGINEERING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

|       | Unit – I                                                                                                              | Marks | BT* | CO* | PO*       |
|-------|-----------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----------|
| 1. a) | Write short notes on:                                                                                                 |       |     |     |           |
|       | i) Detrimental effects of fossil fuel sources                                                                         | 5+5   | L*2 | 1   | 1,2,7     |
|       | ii) Classification of Biofuels, based on different generations                                                        |       |     |     |           |
| b)    | Explain Algal biomass as an attractive source for biofuel production.                                                 | 10    | L2  | 2   | 1,2,7     |
| 2. a) | Describe the biofuel production with Starch tubers Cassava (tapioca) and sweet potato as sources.                     | 10    | L2  | 1   | 1,2,7     |
| b)    | Describe lipase catalyzed inter-esterification and transesterification.                                               | 10    | L2  | 2   | 1,2,7     |
| 3. a) | Explain the importance of Steam explosion and Alkali pre-treatment methods for breakdown of lignocellulosic material. | 10    | L2  | 1   | 1,2,7     |
| b)    | Explain the process of batch and fed batch fermentation for the production of ethanol.                                | 10    | L2  | 2   | 1,2,7     |
|       | <b>Unit – II</b>                                                                                                      |       |     |     |           |
| 4. a) | Describe the factors that play a major role in hydrogen production.                                                   | 10    | L2  | 3   | 1,2,5,7   |
| b)    | Discuss the limitations of following reactors:                                                                        |       |     |     |           |
|       | (i) Integrated continuous stirred tank reactors                                                                       |       |     |     |           |
|       | (ii) Anaerobic fluidized bed reactor                                                                                  | 5+5   | L2  | 4   | 1,2,5,6,7 |
| 5. a) | With illustration explain:                                                                                            |       |     |     |           |
|       | (i) Single chamber MFC                                                                                                | 5+5   | L3  | 3   | 1,2,5,7   |
|       | (ii) Two chamber MFC                                                                                                  |       |     |     |           |
| b)    | Give an account on feedstock used for biohydrogen production.                                                         | 10    | L2  | 4   | 1,2,5,6,7 |
| 6. a) | Discuss the production of hydrogen by batch fermentation process.                                                     | 10    | L3  | 3   | 1,2,5,7   |
| b)    | How do you measure the performance of MFC by adopting substrate and biomass measurement?                              | 10    | L2  | 4   | 1,2,5,6,7 |
|       | <b>Unit – III</b>                                                                                                     |       |     |     |           |
| 7. a) | Write a note on Biogas digester models available in India.                                                            | 10    | L2  | 5   | 1,2,5,6,7 |
| b)    | Write short notes on following:                                                                                       |       |     |     |           |
|       | (i) Refuse derived fuel (RDF) system                                                                                  |       |     |     |           |
|       | (ii) Fluidized bed incineration                                                                                       | 5+5   | L2  | 5   | 1,2,5,6,7 |
| 8. a) | Give a brief account on anaerobic processing of waste and types of digesters.                                         | 10    | L2  | 5   | 1,2,5,6,7 |
| b)    | Describe the hydrothermal processing technology.                                                                      | 10    | L2  | 5   | 1,2,5,6,7 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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